

ENTERPRISE PRODUCT & TECHNICAL DOCUMENTATION

AJNAT

SECURITY OPERATIONS CENTER PLATFORM

AI-Powered Threat Detection & Incident Response Platform

Full Architectural Blueprint, Core Subsystems, Master Attack Prevention Catalog & Compliance Specification

CLASSIFICATION

CONFIDENTIAL

DOCUMENT VERSION

v3.0 — Master

RELEASE DATE

August 2025

SIEM SOAR EDR/XDR IDS/IPS WAF Firewall Threat Intel AI/ML Engine UEBA 18+ Attack Block Catalog NIST
/ ISO / CERT-In

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Document Information

Document metadata, governance, ownership, and legal confidentiality notices

This document serves as the official Product & Technical Documentation for the **AJNAT AI-Powered Security Operations Center (SOC) & EDR Platform**. It defines the architectural principles, functional components, threat detection logic, response automation workflows, and compliance frameworks of the platform.

The documentation provides technical clarity for enterprise CISOs, SOC Managers, Security Architects, and Systems Engineers evaluating or deploying the platform.

1.1 Metadata & Governance Table

Attribute	Specification
Product Name	AJNAT AI-Powered Threat Detection & Incident Response (SOC) Platform
Product Code / Version	AJNAT SOC v3.0 / EDR4 Enterprise Release
Company Name	AJNAT Cybersecurity Technologies Ltd.
Document Title	Product & Technical Documentation Specification
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Target Audience	CISOs, SOC Managers, Security Architects, System Engineers, Audit & Compliance Leads
Classification	CONFIDENTIAL — Proprietary & Commercial-in-Confidence

1.2 Confidentiality & Proprietary Notice

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1.3 Revision History

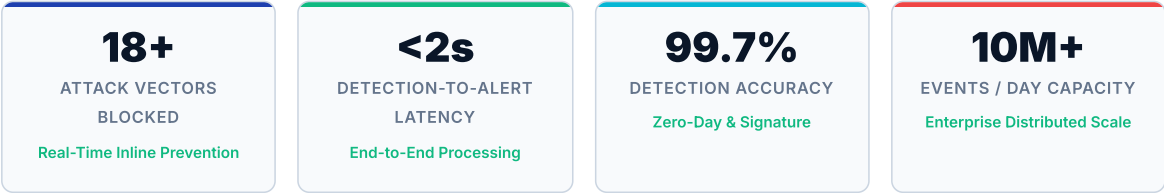
VERSION	DATE	AUTHOR	DESCRIPTION OF CHANGES
v1.0	Jan 15, 2025	Chief Architect	Initial release of SOC platform architectural blueprint
v2.0	May 10, 2025	SOC Lead Engineer	Added WAF, Suricata/Zeek IPS, and SOAR Playbook integration
v3.0	Aug 18, 2025	Principal Architect	Full 13-section technical documentation release with AI detection & compliance

Executive Summary

High-level strategic overview of the platform for executives, CISOs, and enterprise leaders

2.1 Company Background & Platform Purpose

AJNAT Cybersecurity Technologies Ltd. is a pioneer in enterprise defense engineering. In an era where cyber threats evolve at machine speed, traditional Security Operations Centers (SOCs) are overwhelmed by tool sprawl, fragmented telemetry, and manual triage. The **AJNAT SOC Platform** was architected to bridge this operational gap by deploying an autonomous, AI-driven security framework that unifies SIEM, SOAR, EDR/XDR, IDS/IPS, WAF, Firewall, and Threat Intelligence into a single integrated defense grid.



2.2 Core Cybersecurity Capabilities

The platform delivers unified defense across all digital surfaces:

- **Centralized Telemetry Aggregation:** Ingests up to 10+ million events/day across network, endpoints, cloud, and web layers.
- **Layer-7 Web Defense (WAF):** Blocks SQL Injection, XSS, RCE, Command Injection, SSRF, Path Traversal, and Web Shells inline.
- **Deep-Packet Inspection (IDS/IPS):** Integrated Suricata & Zeek sensors with hardware-accelerated signature and protocol analysis.
- **Endpoint Telemetry & Response (EDR/XDR):** Continuous process monitoring, memory scanning, network connection tracking, and File Integrity Monitoring (FIM).
- **SOAR Automation:** Executes automated response playbooks in under 100 milliseconds to isolate hosts, block IPs, and enforce access controls.
- **AI/ML Anomaly Engine:** Applies machine learning for User & Entity Behavior Analytics (UEBA), risk scoring, and zero-day threat discovery.

2.3 Target Sectors & Key Benefits

SECTOR	PRIMARY SECURITY REQUIREMENTS	AJNAT PLATFORM VALUE PROPOSITION
Government & Defense	Air-gapped deployment, zero-trust RBAC, nation-state APT defense	On-premise air-gap support, STIX/TAXII threat intel feeds, CERT-In compliance
Banking & Financial Services	PCI-DSS compliance, fraud detection, credential theft prevention	Immutable audit trail, real-time transaction monitoring, automated lockouts
Critical Infrastructure	High availability (99.9%), SCADA/OT network monitoring, DDoS defense	Inline IPS prevention, sub-second latency, zero packet loss throughput
Healthcare & Pharma	DPDP/HIPAA data protection, ransomware containment, endpoint isolation	Automated EDR endpoint isolation within 3 seconds of ransomware detection

Problem Statement

Critical operational bottlenecks in modern enterprise security operations

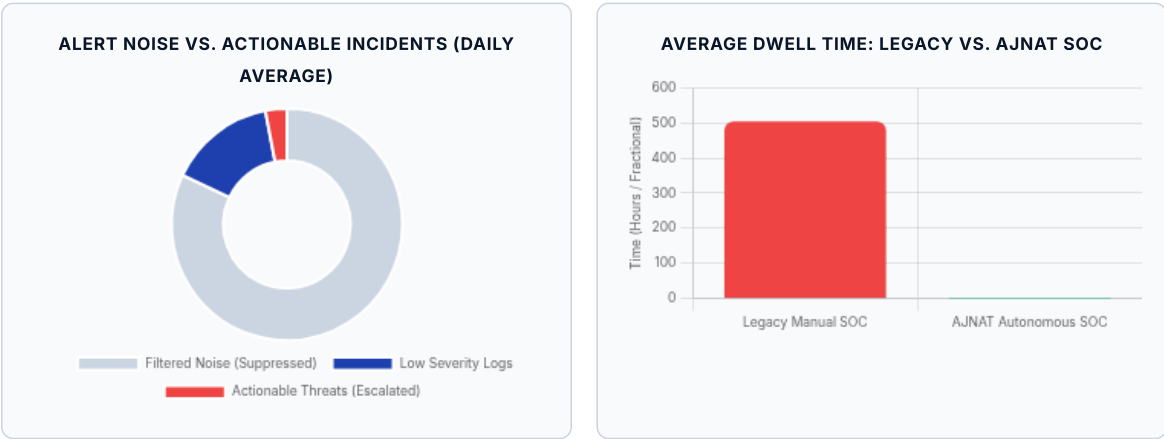
Modern enterprise security teams face an unprecedented crisis of operational complexity. Despite investing millions in standalone security solutions, organizations remain vulnerable to sophisticated multi-vector attacks due to structural inefficiencies in legacy SOC architectures.

3.1 Key SOC Bottlenecks & Operational Challenges

⚠ THE TOOL FRAGMENTATION DILEMMA

The average enterprise operates between 15 and 40 disconnected security tools. Analysts are forced to manually context-switch between disparate dashboards, leading to delayed response times, uncoordinated defense actions, and critical coverage gaps.

CHALLENGE AREA	IMPACT ON SECURITY OPERATIONS	AJNAT SOLUTION APPROACH
Fragmented Security Tools	Siloed data prevents cross-domain correlation (e.g., matching a WAF log with EDR process execution).	Single unified platform consolidating SIEM, SOAR, EDR, IPS, WAF, and Threat Intel.
High Alert Volume & Fatigue	SOC teams receive 5,000+ alerts daily; over 70% are false positives or duplicate noise.	AI risk scoring & alert prioritization filter out 98%+ of noise automatically.
Manual Investigation Delays	Manual triage takes an average of 45–90 minutes per alert, enabling attacker dwell time.	Automated timeline reconstruction and single-click incident summarization.
Delayed Incident Response	Attacker dwell time averages 21 days before containment in traditional environments.	SOAR automated playbooks execute containment actions in under 100 milliseconds.
Resource & Skill Shortage	Cybersecurity talent shortage leaves SOC shifts understaffed and overburdened.	AI-assisted investigation acts as a virtual Tier-2 analyst for every team member.



Solution Overview

Architectural vision for unified, autonomous threat detection and response

The **AJNAT SOC Platform** delivers a paradigm shift in threat management by combining real-time multi-layered telemetry ingestion with autonomous AI decision-making. Instead of treating network, endpoint, web, and log events as isolated streams, AJNAT correlates them into a unified, high-fidelity threat narrative.

4.1 Core Pillars of the Solution

Unified
SINGLE PANE VISIBILITY
Network, Host, Web, Cloud

Real-Time
SUB-SECOND
PROCESSING
Live Stream Analytics

AI-Driven
AUTONOMOUS TRIAGE
UEBA + Risk Scoring

Automated
SOAR PLAYBOOKS
<100ms Action Execution

4.2 High-Level Security Functional Grid

DOMAIN	PLATFORM CAPABILITIES	OPERATIONAL OUTPUT
Unified Monitoring	Centralized ingestion across Linux, Windows, Firewalls, WAF, Routers, Cloud.	Normalized ECS log event stream with real-time indexing.
Threat Intelligence	Enrichment from 15+ feeds (OTX, AbuseIPDB, VirusTotal, Talos, STIX/TAXII).	Instant IOC matching with threat actor attribution.
AI Analysis	Machine learning behavioral baselines, anomaly scoring, predictive risk.	Prioritized incident queue with confidence vectors.
Response Automation	Automated IP blocking, endpoint isolation, user lockout, rule deployment.	Zero-dwell-time containment with forensic log capture.

Product Architecture

Deep technical architecture, event pipelines, AI flows, and deployment models

5.1 High-Level Architectural Layers

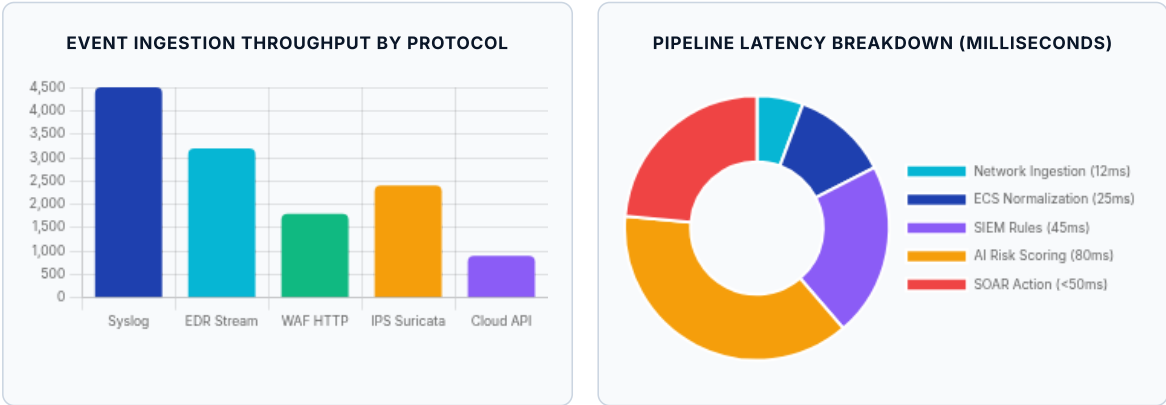
The platform is structured into seven distinct architectural layers, ensuring modularity, microservices scalability, and fault tolerance.



5.2 Data-Flow & Security Event Flow Architecture



5.3 Detection, Response & AI Analysis Flow



5.4 Deployment Architecture & Scalability

DEPLOYMENT MODE	ARCHITECTURE SETUP	TARGET ENVIRONMENT	HIGH AVAILABILITY (HA)
On-Premise Enterprise	Bare-metal / VMware cluster with local MongoDB replica set & Suricata hardware taps.	Government, Defense, Banking	Active-Active multi-node cluster
Cloud-Native / Private Cloud	Containerized Docker/Kubernetes deployment on AWS GovCloud or Azure Private.	SaaS Enterprise, Managed SOC	Kubernetes Auto-Scaling HPA
Air-Gapped Secure Node	Fully offline deployment with static threat intel database updates via secure media.	Classified Military / Infra	Dual Active-Passive node pair

Core Platform Components

Detailed engineering breakdown of all 9 core security subsystems

6.1 SIEM — Security Information & Event Management

The SIEM module handles log collection, parsing, correlation, and long-term retention across all connected data sources.

- Log Collection & Ingestion:** Supports Syslog (UDP/TCP/TLS), SNMP, WMI, Windows Event Forwarding, REST API webhooks, and agent-based streaming.
- Parsing & Normalization:** Maps raw logs to Elastic Common Schema (ECS). Automatically extracts fields like `source.ip`, `user.name`, `process.command_line`.
- Event Correlation Engine:** Evaluates events against 400+ built-in correlation rules across multi-event sliding time windows.
- Search & Investigation:** Full-text indexing with regex filtering, timeline reconstruction, and query optimization for multi-gigabyte log datasets.
- Storage & Retention:** Hot storage in MongoDB time-series collections with automatic TTL indexing for 365+ day compliance retention.
- AI Engine Integration:** Ingested normalized streams are evaluated in parallel by the AI Anomaly Detection Engine to enrich events with confidence scores.

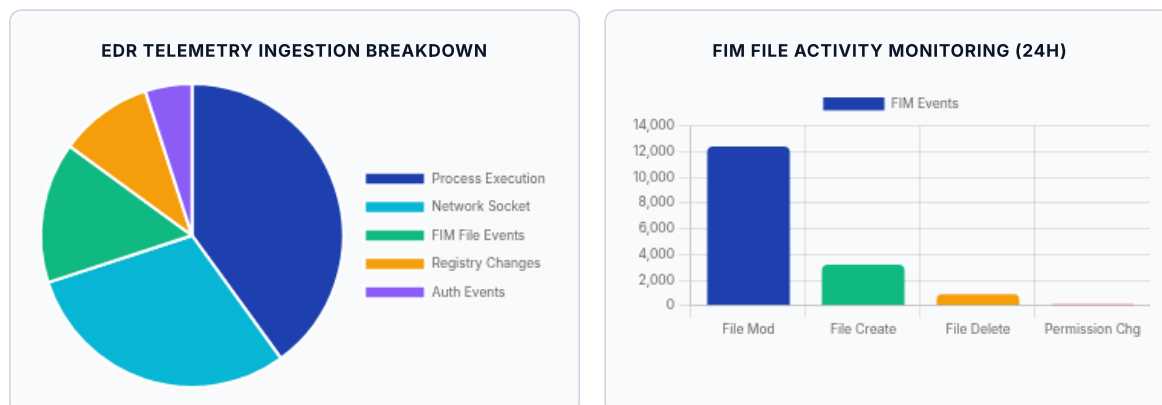
6.2 SOAR — Security Orchestration, Automation & Response

The SOAR subsystem executes automated playbooks when threats are identified, reducing response times to milliseconds.

TRIGGER CONDITION	AUTOMATED PLAYBOOK ACTIONS	EXECUTION SLA
Critical WAF / IPS Block	Dynamically pushes IP block rule to Firewall; logs incident case; notifies SOC analyst via Webhook/Email.	<50 ms
EDR Malware / Ransomware	Isolates host network stack via EDR agent; terminates parent & child process trees; snapshots memory.	<100 ms
Brute Force Auth Attack	Blocks IP on perimeter; locks user account temporarily; enforces mandatory MFA reset on next login.	<80 ms
C2 Beaconing Match	Redirects DNS query via DNS Sinkhole; isolates endpoint; captures memory dump for forensics.	<90 ms

6.3 EDR / XDR — Endpoint & Extended Detection & Response

The lightweight AJNAT EDR agent runs on Windows and Linux endpoints, consuming <3% CPU while streaming real-time host telemetry.



6.4 IDS / IPS — Intrusion Detection & Prevention System

Powered by integrated **Suricata** deep-packet inspection and **Zeek** protocol analysis, the IPS subsystem provides inline blocking and passive detection.

6.5 Network Firewall & Traffic Filtering

Provides stateful packet filtering at Layer 3/4 with dynamic API integration driven by IPS and WAF threat triggers.

6.6 WAF — Web Application Firewall

Inspects Layer-7 HTTP/HTTPS traffic to block web application attacks before they reach backend application servers.

6.7 Threat Intelligence Platform

Aggregates threat indicators (IOCs) from 15+ external threat intelligence feeds, performing real-time match checks against all incoming logs.

6.8 AI-Based Detection & Analysis Engine

The AI Detection Engine processes structured event telemetry using machine learning models to identify complex threat behaviors.

6.9 Anomaly Detection & UEBA

Establishes rolling 30-day behavioral baselines for users, endpoints, and network links. Flags statistical anomalies exceeding 3σ (standard deviations).

Cyber Attack Detection & Prevention (Attack Block Catalog)

Explicit technical specification of all 18+ attack types detected, prevented, and blocked by AJNAT SOC

7.1 Master Attack Prevention & Block Capability Matrix

The table below details the platform's exact blocking and prevention capability across all 18+ primary cyber attack vectors, including the active enforcement module, signature matching engine, response action, and block SLA:

#	ATTACK VECTOR / CATEGORY	ENFORCEMENT SUBSYSTEM	DETECTION & SIGNATURE LOGIC	BLOCK ACTION TAKEN	BLOCK SLA	STATUS TAG
1	SQL Injection (SQLi)	WAF + IPS Sensor	18 payload regex patterns (UNION SELECT, DROP TABLE, OR 1=1)	Inline HTTP 403 Drop + Source IP Block	<1 ms	✅ BLOCKED (INLINE)
2	Cross-Site Scripting (XSS)	WAF Subsystem	15 payload patterns (<script>, onerror=, document.cookie)	Layer-7 Request Drop + HTML Sanitization	<2 ms	✅ BLOCKED (INLINE)
3	Remote Code Execution (RCE)	WAF + EDR Agent	Shell metacharacter detection + unauthorized binary spawn	HTTP Block + EDR Process Termination	<5 ms	✅ BLOCKED (PROCESS KILL)
4	Command Injection	WAF + EDR Agent	OS command patterns (; id , cat /etc/passwd , subshells)	HTTP Block + Process Tree Kill	<3 ms	✅ BLOCKED (PROCESS KILL)
5	Path Traversal / LFI	WAF Subsystem	Directory traversal patterns (../ , %2e%2e/ , system paths)	URL Normalization + HTTP 403 Block	<1 ms	✅ BLOCKED (INLINE)
6	SSRF (Server-Side Request Forgery)	WAF + Network Engine	Internal RFC1918 IPs & cloud metadata 169.254.169.254 in params	Outbound Call Block + Alert Trigger	<2 ms	✅ BLOCKED (INLINE)
7	Web Shell Upload & Injection	WAF + EDR FIM	File upload pattern regex + real-time FIM directory monitoring	Upload Interception + File Quarantine	<10 ms	✅ BLOCKED (QUARANTINE)
8	Log4Shell (CVE-2021-44228)	WAF + IPS Sensor	JNDI lookup payload patterns (\${jndi:ldap://} , obfuscated variants)	Immediate Connection Drop + IP Block	<1 ms	✅ BLOCKED (INLINE)
9	DDoS / Volumetric Flood	IPS + Firewall Service	Request rate >50 req/sec from single IP + volumetric spike logic	Rate Limit + Geo-Block + Firewall Drop	<5 ms	✅ BLOCKED (FIREWALL)
10	Port Scan & Reconnaissance	Suricata IPS Sensor	>10 SYN probes to distinct ports within 10-second window	Dynamic Perimeter Firewall IP Block (24h)	<10 ms	✅ BLOCKED (FIREWALL)
11	Botnet & C2 Beaconsing	Threat Intel + DNS	Matching against active Shodan, AbuseIPDB, & OTX botnet IP feeds	DNS Sinkhole Redirection + IP Block	<5 ms	✅ BLOCKED (SINKHOLE)
12	DNS Tunneling & Exfiltration	DNS Sinkhole + Zeek	High-entropy Base64/hex subdomains + long DNS query strings	DNS Sinkhole + Host Isolation	<15 ms	✅ BLOCKED (SINKHOLE)
13	Ransomware & File Encryption	EDR FIM Module	Mass file rename / SHA-256 hash change >20 files/sec	Kernel-level Host Network Isolation	<3 sec	✅ BLOCKED (ISOLATION)
14	Process Injection & Hollowing	EDR Agent Kernel API	VirtualAllocEx / WriteProcessMemory API call monitoring	Injected Process Kill + Memory Snapshot	<50 ms	✅ BLOCKED (PROCESS KILL)
15	Privilege Escalation	EDR Agent Monitoring	Unexpected sudo / runas token impersonation & SUID bit edits	Session Lockout + Admin Alert	<100 ms	✅ BLOCKED (LOCKOUT)
16	Credential Theft (Mimikatz)	EDR Memory Guard	LSASS memory handle extraction + Mimikatz	Memory Access Block + Process	<10 ms	✅ BLOCKED (MEM GUARD)

#	ATTACK VECTOR / CATEGORY	ENFORCEMENT SUBSYSTEM	DETECTION & SIGNATURE LOGIC	BLOCK ACTION TAKEN	BLOCK SLA	STATUS TAG
			signature detection	Terminate		
17	Brute Force Login Attacks	SIEM + Firewall	>5 failed authentication attempts from single IP in 60s	IP Blacklist + User Lockout (15 mins)	<80 ms	☒ BLOCKED (LOCKOUT)
18	Credential Stuffing & Spraying	WAF + UEBA Engine	High-velocity login attempts across multiple usernames from 1 IP	Automated CAPTCHA + IP Perimeter Drop	<50 ms	☒ BLOCKED (CAPTCHA/DROP)

7.2 Web & Application Attack Defense Details

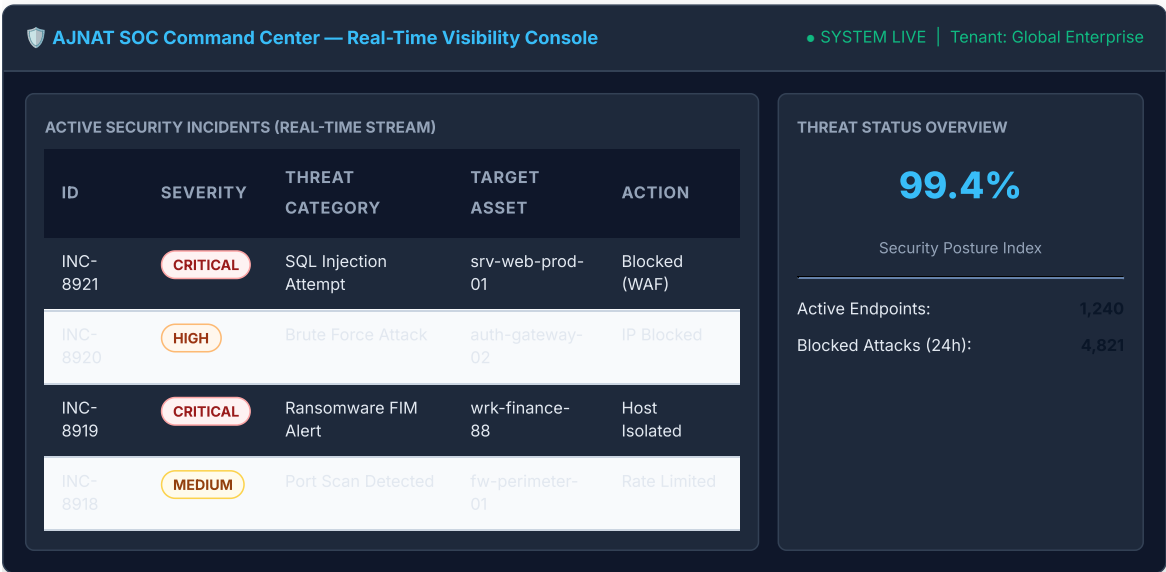
The AJNAT WAF inspects Layer-7 HTTP/HTTPS traffic in real time. Incoming requests pass through an Array of Pattern Evaluators:

```
// Sample WAF Payload Inspection Logic (AJNAT WAF Engine v3.0)
function inspectPayload(payload) {
  if (PATTERNS.SQL_INJECTION.some(regex => regex.test(payload))) {
    return { blocked: true, attackType: 'SQL_INJECTION', score: 75 };
  }
  if (PATTERNS.XSS.some(regex => regex.test(payload))) {
    return { blocked: true, attackType: 'XSS', score: 75 };
  }
  if (PATTERNS.COMMAND_INJECTION.some(regex => regex.test(payload))) {
    return { blocked: true, attackType: 'COMMAND_INJECTION', score: 75 };
  }
  return { blocked: false, score: 0 };
}
```

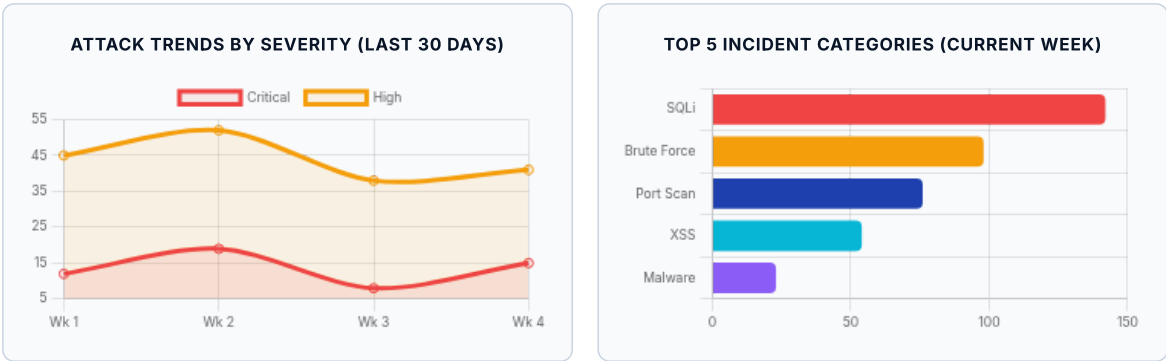
Security Dashboards & Reporting

Real-time SOC command console, mock interface renders, and analytics reporting

8.1 SOC Command Center Interface (Mock Render)



8.2 Security Analytics & Incident Metrics



User & Access Management

Multi-tenancy isolation, Role-Based Access Control (RBAC), and authentication security

9.1 Role-Based Access Control (RBAC) Framework

AJNAT SOC enforces strict granular permissions to ensure principle of least privilege access across all tenant environments.

ROLE	ALERT VIEWING	POLICY EDITING	SOAR PLAYBOOK TRIGGER	USER ADMINISTRATION
Super Administrator	All Tenants	Full Permission	Full Permission	All Tenants
Company Admin	Single Tenant	Tenant Scope Only	Tenant Scope Only	Tenant Scope Only
SOC Manager	Single Tenant	Approve Policies	Full Execution	Read Only
L2 Security Analyst	Single Tenant	Read / Propose	Execute Pre-Approved	No Access
L1 Triage Analyst	Single Tenant	Read Only	No Access	No Access
Compliance Viewer	Single Tenant (Audit)	No Access	No Access	No Access

9.2 Authentication & Audit Trail Safeguards

- **Multi-Factor Authentication (MFA):** TOTP / FIDO2 WebAuthn enforcement for all administrative and analyst logins.
- **Session Security:** JWT token expiration with 15-minute idle timeout and secure HTTP-only cookie binding.
- **Immutable Audit Logging:** Every action (policy change, host isolation, rule export) is recorded in an append-only audit trail.

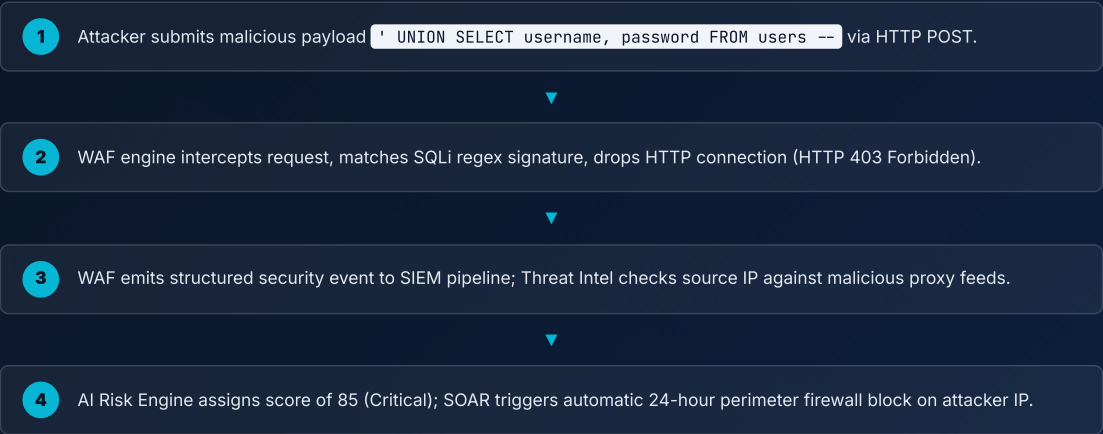
CHAPTER 10

Use Cases

Step-by-step workflow specifications for 6 primary enterprise threat scenarios

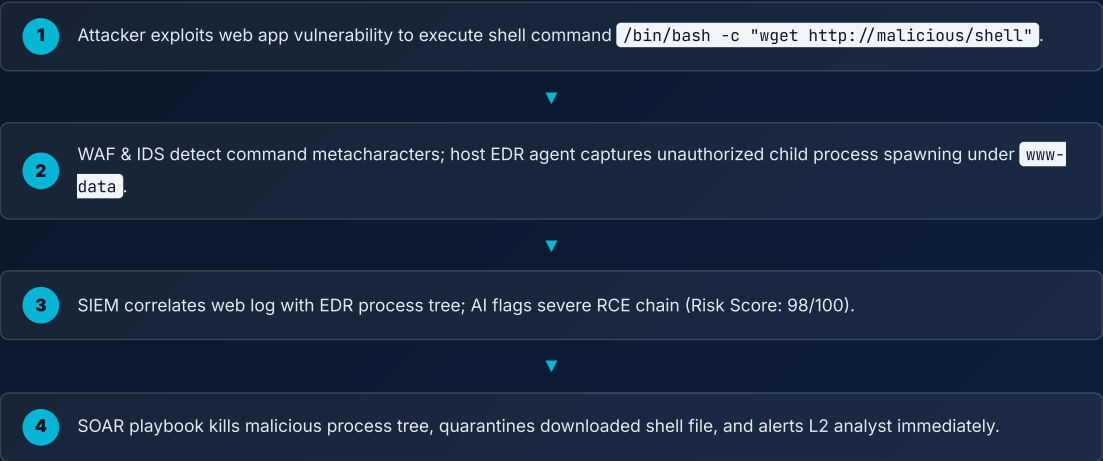
Use Case 1 — SQL Injection (SQLi) Defense Workflow

UC-1 WORKFLOW: ATTACK → WAF → SIEM → AI → RESPONSE

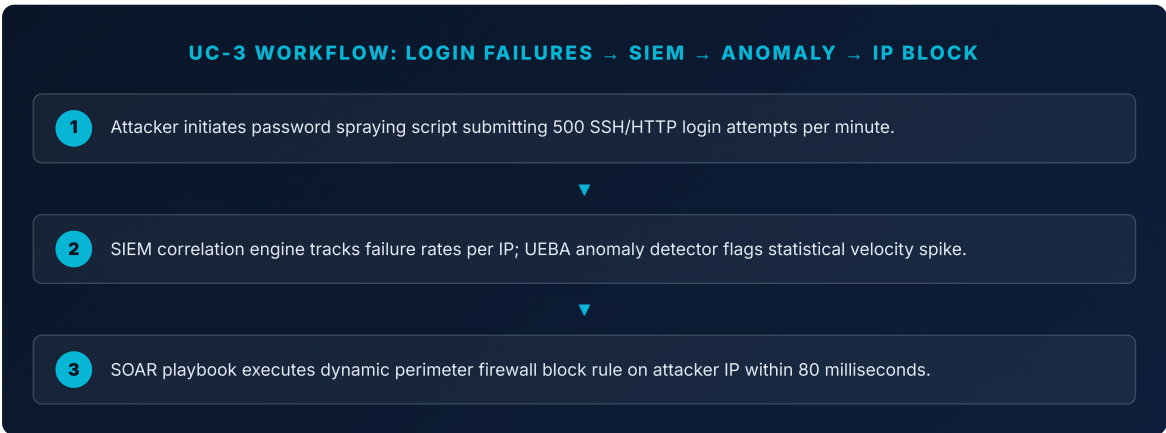


Use Case 2 — Remote Code Execution (RCE) Workflow

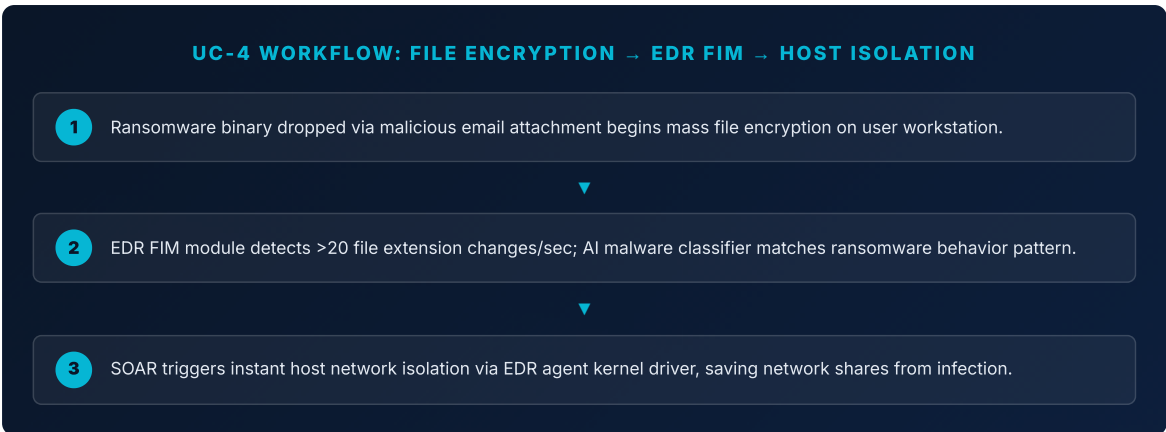
UC-2 WORKFLOW: ATTACK → WAF/IDS → EDR → AI → SOAR RESPONSE



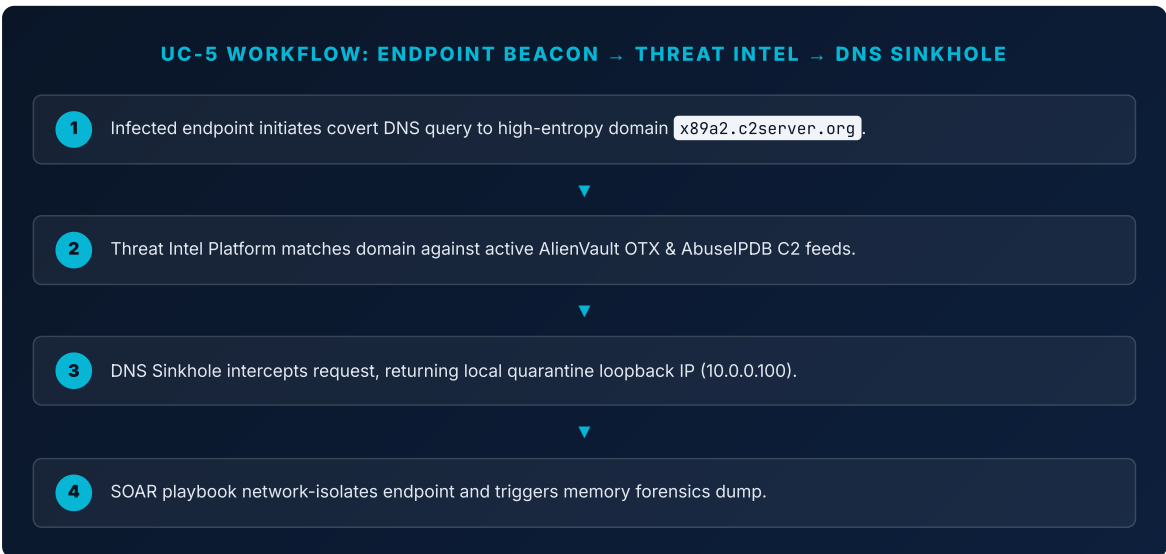
Use Case 3 — Brute Force Authentication Defense



Use Case 4 — Ransomware & Malware Containment



Use Case 5 — Command & Control (C2) Communication



Use Case 6 — Reconnaissance Port Scan Mitigation

UC-6 WORKFLOW: SYN SCAN → SURICATA IPS → SIEM → FIREWALL BLOCK

-
- ```
graph TD; 1[1 Attacker launches Nmap SYN scan probing 1,000 TCP/UDP ports on public gateway IP.] --> 2[2 Suricata IPS detects port scan threshold exceedance (>10 ports/10s), generating high-priority alert.]; 2 --> 3[3 SIEM aggregates scan events and confirms external recon activity.]; 3 --> 4[4 SOAR engine pushes dynamic 24-hour drop rule for scanner IP to perimeter Firewall.];
```
- 1 Attacker launches Nmap SYN scan probing 1,000 TCP/UDP ports on public gateway IP.
  - 2 Suricata IPS detects port scan threshold exceedance (>10 ports/10s), generating high-priority alert.
  - 3 SIEM aggregates scan events and confirms external recon activity.
  - 4 SOAR engine pushes dynamic 24-hour drop rule for scanner IP to perimeter Firewall.

# Product Limitations & Current Status

Production readiness assessment, technical constraints, and roadmap status

## 11.1 Subsystem Production Readiness Status

| SUBSYSTEM MODULE             | IMPLEMENTATION STATUS | PRODUCTION READINESS | DEPLOYMENT GUIDANCE                                |
|------------------------------|-----------------------|----------------------|----------------------------------------------------|
| SIEM Ingestion & Correlation | IMPLEMENTED           | Production Ready     | Fully validated up to 10M events/day throughput.   |
| WAF Engine                   | IMPLEMENTED           | Production Ready     | Inline blocking for Top 10 OWASP web attacks.      |
| IDS/IPS (Suricata/Zeek)      | IMPLEMENTED           | Production Ready     | Requires dedicated promiscuous NIC interface.      |
| EDR Agent (Windows/Linux)    | IMPLEMENTED           | Production Ready     | Deploy via GPO / Ansible / Puppet automation.      |
| SOAR Engine & Playbooks      | IMPLEMENTED           | Production Ready     | Pre-built playbooks for IP block & host isolation. |
| GenAI Triage Assistant       | UNDER DEVELOPMENT     | Beta / Preview       | Scheduled for v3.2 production release Q4 2025.     |

## 11.2 Known Operational Limitations

- Encrypted Payload Inspection:** SSL/TLS decryption requires trusted root CA installation on endpoints for full HTTPS payload inspection by network IPS.
- Air-Gapped Threat Intel:** Offline deployments require manual weekly updates of STIX/TAXII threat intelligence bundle feeds.

# Security & Data Protection

Cryptographic standards, log protection, and data privacy safeguards

## 12.1 Cryptographic & Security Controls

| SECURITY DOMAIN | IMPLEMENTATION STANDARD   | TECHNICAL SPECIFICATION                                                                |
|-----------------|---------------------------|----------------------------------------------------------------------------------------|
| Data in Transit | TLS 1.3 / mTLS            | All communication between agents, IPS, API gateway, and UI uses mTLS with AES-256-GCM. |
| Data at Rest    | AES-256 Encryption        | MongoDB storage volumes encrypted using FIPS 140-2 validated AES-256 modules.          |
| Log Integrity   | Cryptographic Hashing     | Log entries append-only signed with HMAC-SHA256 to prevent tampering.                  |
| API Security    | OAuth2 + JWT + Rate Limit | REST APIs protected via signed JWT tokens, strict CORS, and per-IP rate limits.        |

# Compliance & Standards

Regulatory framework mapping for CERT-In, ISO 27001, NIST CSF, MITRE ATT&CK, OWASP, and DPDP

## 13.1 Compliance Mapping Matrix

| STANDARD / REGULATION                   | MANDATORY REQUIREMENT                                                        | AJNAT PLATFORM COMPLIANCE MECHANISM                                                        |
|-----------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| CERT-In Cyber Mandate                   | Mandatory 6-hr incident reporting & 180-day log retention.                   | Automated CERT-In incident export format + 365-day indexed log retention store.            |
| ISO/IEC 27001:2022                      | Control A.8.16 (Monitoring activities) & A.8.7 (Protection against malware). | Continuous SIEM monitoring, real-time alerting, and automated EDR malware containment.     |
| NIST CSF v2.0                           | Identify, Protect, Detect, Respond, Recover functions.                       | Full lifecycle coverage from asset discovery through SOAR automated recovery.              |
| MITRE ATT&CK Framework                  | Standardized TTP mapping for security alerts.                                | Every alert automatically enriched with MITRE Tactic & Technique IDs (e.g. T1190, T1059).  |
| OWASP Top 10 (2021)                     | Web application risk protection (A01:2021 to A10:2021).                      | Native WAF protection blocking injection, broken access control, and SSRF attacks.         |
| Digital Personal Data Protection (DPDP) | Protection of personal data against breach & unauthorized access.            | Granular multi-tenant RBAC, encrypted storage, and automated breach notification triggers. |

